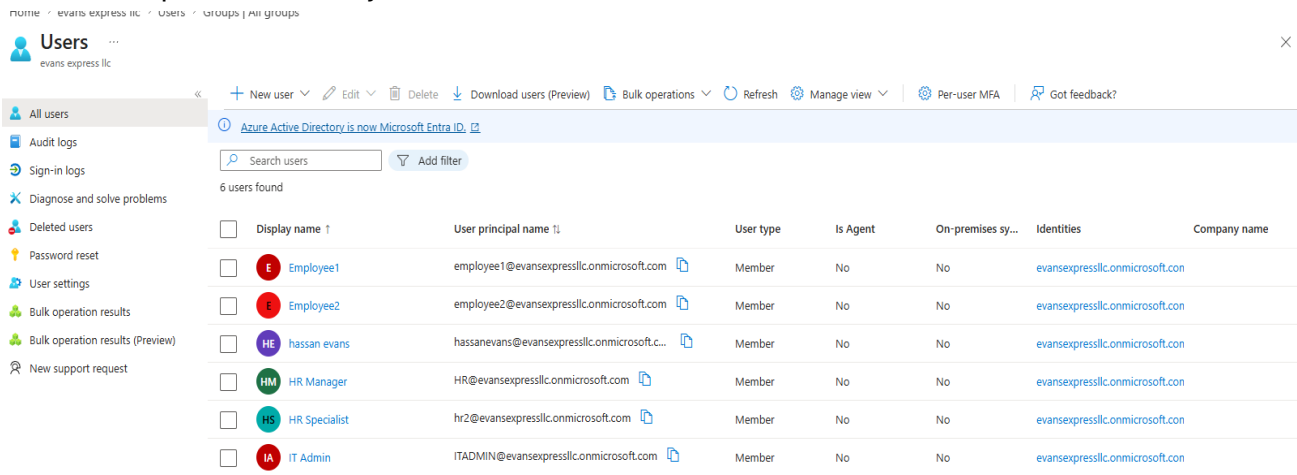


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This project focused on designing and implementing an Identity and Access Management (IAM) environment using Microsoft Entra ID to simulate a real-world enterprise security setup. The project was structured across three phases. I will explain each phase that took place and I will also show screenshots of the environment to show my work.

In the first phase, user accounts and security groups were created to represent different departments and job roles.



The screenshot shows the Microsoft Entra ID 'Users' page. The left sidebar contains navigation links: All users, Audit logs, Sign-in logs, Diagnose and solve problems, Deleted users, Password reset, User settings, Bulk operation results, Bulk operation results (Preview), and New support request. The main area displays a table of 6 users found. The table has columns for checkboxes, Display name, User principal name, User type, Is Agent, On-premises sys..., Identities, and Company name. The users listed are Employee1, Employee2, hassan evans, HR Manager, HR Specialist, and IT Admin.

☐	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sys...	Identities	Company name
☐	Employee1	employee1@evansexpressilc.onmicrosoft.com	Member	No	No	evansexpressilc.onmicrosoft.com	
☐	Employee2	employee2@evansexpressilc.onmicrosoft.com	Member	No	No	evansexpressilc.onmicrosoft.com	
☐	hassan evans	hassanevans@evansexpressilc.onmicrosoft.c...	Member	No	No	evansexpressilc.onmicrosoft.com	
☐	HR Manager	HR@evansexpressilc.onmicrosoft.com	Member	No	No	evansexpressilc.onmicrosoft.com	
☐	HR Specialist	hr2@evansexpressilc.onmicrosoft.com	Member	No	No	evansexpressilc.onmicrosoft.com	
☐	IT Admin	ITADMIN@evansexpressilc.onmicrosoft.com	Member	No	No	evansexpressilc.onmicrosoft.com	

Five user accounts were created to represent a cross-section of roles within the organization: two general employees, two HR specialists, one HR manager, and one IT administrator.

Each user was then assigned to one or more security groups based on their role. Three groups were configured: an Employee group, which all users were added to as a baseline; an HR group, which included the HR specialists and HR manager; and an IT group, which included the IT administrator. This structure ensured that access and permissions could be managed at the group level rather than on a per-user basis, making the environment easier to maintain and scale.

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Home > evans express llc > Users > Groups | All groups > Users > Groups | All groups > IT Administrator

IT Administrator

Group

Members

×

« + Add members Bulk operations Refresh Manage view Remove Got feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

New support request

Direct membersAll members

Search

Add filter

1 group member found

☐	Name ↑	Type	Email	User type	Object Id
☐	IA IT Admin	User		Member	cd0fffc0-dd4f-44f5-a9b1-852d18ebf272

Home > evans express llc > Users > Groups | All groups > Users > Groups | All groups > HR GROUP

HR GROUP

Group

Members

×

« + Add members Bulk operations Refresh Manage view Remove Got feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

New support request

Direct membersAll members

Search

Add filter

2 group members found

☐	Name ↑	Type	Email	User type	Object Id
☐	HM HR Manager	User		Member	09d8c939-b6c9-4f73-96bc-ddc9d567ad7b
☐	HS HR Specialist	User		Member	ff39f565-4246-4f35-ad13-1ba3d41a2a5b

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Home > evans express llc > Users > Groups | All groups > Users > Groups | All groups > All Employees

All Employees | Members

Group

Overview
Diagnose and solve problems

Manage
Properties
Members
Owners
Roles and administrators
Administrative units
Group memberships
Applications
Licenses
Azure role assignments

Activity
Privileged Identity Management
Access reviews
Audit logs
Bulk operation results
Troubleshooting + Support
New support request

+ Add members Bulk operations Refresh Manage view Remove Got feedback?

Direct members All members

Search Add filter

6 group members found

☐	Name ↑	Type	Email	User type	Object id
☐	 Employee1	User		Member	4b15045e-e0e9-4908-8015-d654bc387ea5
☐	 Employee2	User		Member	ce90114f-85f4-4b33-8fa5-ba7a7d9c3505
☐	 hassan evans	User		Member	a5c40647-0f61-4902-8d82-92504976619e
☐	 HR Manager	User		Member	09d8c939-b6c9-4f73-96bc-ddc9d567ad7b
☐	 HR Specialist	User		Member	ff39f565-4246-4f35-ad13-1ba3d41a2a5b
☐	 IT Admin	User		Member	cd0fffc0-dd4f-44f5-a9b1-852d18ebf272

Role-Based Access Control (RBAC) was applied to enforce the Principle of Least Privilege, ensuring each user only has access appropriate to their responsibilities.

IT Administrators were assigned the Helpdesk Administrator role. This role allows them to reset passwords for non-administrative users, invalidate refresh tokens, create and manage support requests with Microsoft for Azure and Microsoft 365 services, and monitor service health.

HR Managers were assigned the User Administrator role. This role allows them to manage user accounts, manage group memberships, send guest user invitations, assign and manage licenses, and monitor service health. This role was chosen as it aligns with HR's responsibility of managing employees throughout their lifecycle in the organization — from onboarding, where accounts and licenses need to be created and assigned, to offboarding, where accounts need to be disabled or deleted.

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...

evans express llc

Roles and administrators

All roles

Helpdesk Administrator

Assignments

Roles and administrators

All roles

Helpdesk Administrator

Assignments

Roles and administrators

All roles

Helpdesk Administrator

Helpdesk Administrator

All roles

Assignments

Diagnose and solve problems

Manage

Assignments

Description

Activity

Bulk operation results

Troubleshooting + Support

New support request

+ Add assignments

✕ Remove assignments

⬇ Download assignments

🔄 Refresh

🔗 Manage in PIM

🗨 Got feedback?

You can also assign built-in roles to groups now. [Learn More](#)

Search

Search by name

Type

All

Name	UserName	Type	Scope
☐ IT Admin	ITADMIN@evansexpressllc.onmicrosoft.com	User	Directory

Home

Roles and administrators

All roles

Helpdesk Administrator

Assignments

Users

evans express llc

Roles and administrators

All roles

Helpdesk Administrator

Assignments

Roles

User Administrator

All roles

Assignments

Diagnose and solve problems

Manage

Assignments

Description

Activity

Bulk operation results

Troubleshooting + Support

New support request

+ Add assignments

✕ Remove assignments

⬇ Download assignments

🔄 Refresh

🔗 Manage in PIM

🗨 Got feedback?

You can also assign built-in roles to groups now. [Learn More](#)

Search

Search by name

Type

All

Name	UserName	Type	Scope
☐ HR Manager	HR@evansexpressllc.onmicrosoft.com	User	Directory

Successfully added assignment

Successfully added assignment HR Manager

The second phase centered on Zero Trust security, where Conditional Access policies were configured to continuously verify every access request rather than automatically trusting users, strengthening the overall security posture of the environment. I created 3 policies to help the overall security posture of the environment.

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Home > Conditional Access

Conditional Access | Policies

Microsoft Entra ID

Overview

Policies

Deleted Policies

Phased rollout

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

+ New policy + New policy from template Upload policy file What if Refresh Preview features Got feedback?

Microsoft Entra Conditional Access policies are used to apply access controls to keep your organization secure. [Learn more](#)

Microsoft-managed policies

Policies created 0

User created policies

Policies created 3

Search Add filter

3 out of 3 policies found

Policy name	Created by	State	Alert	Creation date	Modified date
Block Non-US Access	USER	Report-only		5/14/2026, 10:24:3...	5/14/2026, 10:48:5...
Require MFA for Admins	USER	Report-only		5/14/2026, 10:21:0...	
Require MFA for Risky Sign Ins	USER	Report-only		5/14/2026, 10:26:3...	

The first policy was blocking all non-US access.

What it does: This policy blocks any sign-in attempts coming from outside the United States. If a user tries to access the environment from a foreign location, they'll be denied — plain and simple.

Why we implemented it: Zero Trust means we don't automatically trust any access request, no matter where it comes from. Since our organization only operates in the US, there's no good reason a legitimate user would be signing in from another country. Blocking foreign access cuts down our attack surface and keeps overseas threat actors out, even if they somehow get their hands on valid credentials.

Home > Conditional Access | Policies

Block Non-US Access

Conditional Access policy

Delete View policy information View policy impact

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies. [Learn more](#)

Name *

Block Non-US Access

Assignments

Users or agents (Preview)

All users

Target resources

No target resources selected

Network

Any network or location and 1 excluded

Conditions

1 condition selected

Access controls

Grant *

Block access

Session *

0 controls selected

The second policy was MFA for admins.

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What it does: This policy requires administrator accounts to pass MFA before they're granted access. It targets elevated accounts like Helpdesk Administrators and User Administrators.

Why we implemented it: Admin accounts carry elevated privileges, which makes them a prime target for attackers. Under Zero Trust, no one gets a free pass — not even admins. By requiring MFA, we make sure that a compromised password alone isn't enough to get in. An attacker would still need that second factor, which goes a long way in preventing account takeovers.

Home > Conditional Access | Policies

Require MFA for Admins

Conditional Access policy

[Delete](#) [View policy information](#) [View policy impact](#)

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies. [Learn more](#)

Name *

Require MFA for Admins

Assignments

Users or agents (Preview) 

Specific users included

Target resources 

All resources (formerly 'All cloud apps')

Network 

Not configured

Conditions 

0 conditions selected

Access controls

Grant * 

1 control selected

Session * 

Sign-in frequency - Every time

The final policy was MFA for risky sign-ins.

What it does: This policy detects sign-ins that are flagged as risky — things like logins from unfamiliar locations, anonymous IP addresses, or unusual behavior — and requires those users to complete MFA before getting in.

Why we implemented it: Under Zero Trust, every access request needs to be verified, full stop. But rather than blocking risky sign-ins outright, we challenge the user with MFA instead. That way, a legitimate user who's traveling or signing in from a new device can still get access, while keeping unauthorized users out. It's a practical balance between security and usability.

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Home > Conditional Access | Policies

Require MFA for Risky Sign Ins ...

Conditional Access policy

Delete

View policy information

View policy impact

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies.
[Learn more](#)

Name *

Require MFA for Risky Sign Ins

Assignments

Users or agents (Preview)

[All users](#)

Target resources

[No target resources selected](#)

Network

[Not configured](#)

Conditions

[1 condition selected](#)

Access controls

Grant *

[1 control selected](#)

Session *

[0 controls selected](#)

The third phase involved setting up SAML-based Single Sign-On (SSO) through an enterprise application integration in Microsoft Entra ID (Azure AD). This allows users to authenticate once and seamlessly access multiple applications without needing to sign in again.

The configuration included registering a new enterprise application, selecting SAML as the sign-on method, and completing the Basic SAML Configuration with the required Identifier (Entity ID) and Reply URL (ACS URL). Attributes and claims were mapped to pass user information such as email and name through the SAML token. The App Federation Metadata was also located for use in connecting the identity provider to the application.

From there, all three security groups — Employee, HR, and IT — were assigned to the application. This ensures that every user in the organization, regardless of their department, is provisioned access through their respective group. Managing access at the group level means that adding or removing a user from a group automatically updates their application access, without requiring any changes to be made on a per-user basis.

Note on Testing:

SSO testing could not be completed in this environment, as the configuration requires real, active URLs (such as a live ACS/Reply URL and Entity ID endpoint) for the SAML authentication flow to be successfully completed. Since this was set up in a lab environment without a live application backend, the test could not be fully executed. Screenshots have been provided to document each step of the configuration process and demonstrate that the setup was completed correctly.

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Home > Conditional Access | Policies > Require MFA for Admins > App registrations > Evans Cargo Express LLC- SSO- APP | Roles and administrators > Enterprise applications | All applications > Evans Cargo Express LLC- SSO- APP

Evans Cargo Express LLC- SSO- APP | SAML-based Sign-on

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

Security

Conditional Access

Permissions

Token encryption

Activity

Sign-in logs

Usage & insights

Audit logs

Provisioning logs

Access reviews

Upload metadata file

Change single sign-on mode

Test this application

Got feedback?

Set up Single Sign-On with SAML

An SSO implementation based on federation protocols improves security, reliability, and end user experiences and is easier to implement. Choose SAML single sign-on whenever possible for existing applications that do not use OpenID Connect or OAuth. [Learn more.](#)

Read the [configuration guide](#) for help integrating Evans Cargo Express LLC- SSO- APP.

1

Basic SAML Configuration

Identifier (Entity ID)
Reply URL (Assertion Consumer Service URL)
Sign on URL
Relay State (Optional)
Logout URL (Optional)

https://company-ss0-app

https://company-ss0-app/signin

Optional

Optional

Optional

Edit

2

Attributes & Claims

givenname
surname
emailaddress
name
Unique User Identifier

user.givenname

user.surname

user.mail

user.userprincipalname

user.userprincipalname

Edit

3

SAML Certificates

Token signing certificate

Status
Thumbprint
Expiration
Notification Email
App Federation Metadata URL
Certificate (Base64)
Certificate (Raw)

Active

B6EAEF86EEDF57C1F009880BD42F561DEF32306E

5/14/2029, 11:19:36 PM

hassanevans@evansexpressllc.onmicrosoft.com

https://login.microsoftonline.com/2e30ff68-35fa-4...

Download

Download

Edit

Home > Conditional Access | Policies > Require MFA for Admins > App registrations > Evans Cargo Express LLC- SSO- APP | Roles and administrators > Enterprise applications | All applications > E

Evans Cargo Express LLC- SSO- APP | SAML-based Sign-on

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

Security

Conditional Access

Permissions

Token encryption

Activity

Sign-in logs

Usage & insights

Audit logs

Provisioning logs

Access reviews

Upload metadata file

Change single sign-on mode

Test this application

Got feedback?

3

SAML Certificates

Token signing certificate

Status
Thumbprint
Expiration
Notification Email
App Federation Metadata URL
Certificate (Base64)
Certificate (Raw)
Federation Metadata XML

Active

B6EAEF86EEDF57C1F009880BD42F561DEF32306E

5/14/2029, 11:19:36 PM

hassanevans@evansexpressllc.onmicrosoft.com

https://login.microsoftonline.com/2e30ff68-35fa-4...

Download

Download

Download

Edit

4

Set up Evans Cargo Express LLC- SSO- APP

You'll need to configure the application to link with Microsoft Entra ID.

Login URL
Microsoft Entra Identifier
Logout URL

https://login.microsoftonline.com/2e30ff68-35fa-4...

https://sts.windows.net/2e30ff68-35fa-4f84-b52b-...

https://login.microsoftonline.com/2e30ff68-35fa-4...

5

Test single sign-on with Evans Cargo Express LLC- SSO- APP

Test to see if single sign-on is working. Users will need to be added to Users and groups before they can sign in.

Test



Evans Cargo Express LLC- SSO- APP | Users and groups

Enterprise Application



- Overview
- Deployment Plan
- Diagnose and solve problems
- Manage
- Properties
- Owners
- Roles and administrators
- Users and groups
- Single sign-on
- Provisioning
- Application proxy
- Self-service
- Custom security attributes
- Security
- Conditional Access
- Permissions
- Token encryption
- Activity
- Sign-in logs
- Usage & insights
- Audit logs
- Provisioning logs

+ Add user/group

Edit assignment

Remove assignment

Update credential

Refresh

Manage view

Got feedback?

The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

First 200 shown, search all users & groups

	Display name	Object type	Role assigned
☐	AE All Employees	Group	User
☐	HG HR GROUP	Group	User
☐	IA IT Administrator	Group	User